Gold Diggers: How to Get More Medals at the Winter Olympics

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ABSTRACT As the Winter 2026 Olympics passed, Canada walked away from Italy in 11th place with 21 medals. A third of Canada's medals came from skating-based competitions, whereas Norway, with 41 medals, placed the majority of its podium finishes in skiing competitions. Out of the 8 sports, 16 disciplines, and 116 competitions in the Olympics, the total theoretical number of medals that a country could get is roughly 348. Put another way, Canada has a total loss of 327 medals that it could have won in the Winter Olympics in Italy. What are the potential influencing factors to win medals in the Winter Olympics? To answer this question, we used web scraping to collect data on what factors correspond with winning medals. Then, we looked at the potential correlation between athletes' age, height, and weight, to see which is the most correlated of the categories to get a podium finish for each discipline. Furthermore, out of all the competitions in the Winter Olympics, certain disciplines offer more opportunities, such that a country could pool their resources on that specific discipline to improve its chances of getting maximum podium finishes. A time-series analysis of previous Winter Olympic events was conducted to make the correlations more concrete with the findings. These findings could help Canada and Canadian athletes get more medals in future Winter Olympics, securing us as the True North strong and free.

INDEX TERMS correlation, webscraping, Winter Olympics, medal efficiency

1. INTRODUCTION

Economic factors have long been identified as key predictors of Olympic success. W. Andreff found that GDP per capita was more important than population in the Winter Olympics, though the relative importance of these two variables reversed in the Summer Olympics[1]. This distinction is significant because the cost of infrastructure in the Winter Games makes national wealth particularly telling, as wealthier nations can invest in specialized sports programs and equipment[2]. Research at the 2014 Sochi Games found that these organizations combine an internal resource-based analysis of athlete performance and elite sport systems with an external market-based analysis of medal opportunity and competition intensity to determine where to direct funding. Using this two-phase approach, target sports are first identified and then continuously reassessed across an Olympic cycle to optimize resource allocation. This strategic targeting helps explain why smaller, well-organized nations like Norway can outperform larger countries such as Canada,

which may distribute resources less efficiently across a broader team.

Beyond economics strategies, the physical demands of individual winter sports play a huge role in the Olympics. Alpine skiing, one of the most technically demanding Winter Olympic disciplines, illustrates how performance is shaped by a rich combination of physiology, equipment, and environmental factors. World-class alpine ski racers must continuously change their technique in response to varying terrain, field layout, and snow conditions [3]. Recent advances in sensor technology now allow kinematics and kinetics to be monitored during both training and competition. This provides detailed anatomical and physiological data that can be used to develop individualized training and equipment strategies for future Olympic athletes [3]. Together, these findings suggest that Winter Olympic success emerges from a multilayered interaction of country decision-making, access to sports, and the mastery of each discipline.

Now that the Milano Cortina 2026 Winter Olympics now over, Canada ended the competition with 21 medals in 11th place. Despite Canada being covered in snow for a large portion of the year, it did not even make the top 10. Norway won the Olympics despite having more than half the team size of Canada overall, but doubled the medals at 41. Norway, also being a northern country, experiences a significant amount of snow, but nowhere near the levels that Canada gets every year. Albeit Norway does have a higher gdp/capita than Canada by almost double the amount, so funding could play a role. So, Canada, having a larger team and coming from a more snowy environment, is still beaten out by smaller teams with less snow at the Winter Olympics. Therefore, what factors drive medal success in the Winter Olympics, by looking at sports opportunities, athlete characteristics, and national or economic factors?

2. DATA ACQUISITION PROCESS AND DATA SOURCES

A. *DATA ACQUISITION PROCESS*

This paper focuses on using web scraping in Python using a variety of different libraries (BS4, requests, urllib). These were used across the websites in part B just below except for Statista.com, which only one document was downloaded for country short form names.

B. *DATA SOURCES*

Olympedia.org [4]

This single website provides a substantial amount of information on athletes across all of the Olympics, even dating back to the original Olympics in Greece. This will be the basis for all of our Athlete information and physical characteristics. The robots.txt does specify it can be webscrapped.

Worldometer.info [5] [6]

This will provide all of the economic and country information that will be used in the analysis. GDP and population will be taken specifically as the robots.txt allows it.

Statista.com [7]

This website provided a document that had the country short form name that was used to help join tables together.

3. METHODS (LINKING, CLEANING, and ANALYSIS APPROACH)

A. *LINKING DATA*

Data was collected from two external domains and unified in a local SQLite database (olympics.db). Athlete and medal data were scraped from Olympedia.org across five Winter Olympics editions (2010, 2014, 2018, 2022, 2026) using a two-stage traversal. In the first stage, podium.py requested each edition's medal results page (e.g., /editions/57/medal) and parsed the results table to extract athlete

names and profile URLs, storing a deduplicated set of records in the podium table. This constitutes the multi-page traversal, as five separate edition pages were navigated to seed the athlete URL list.

In the second stage, athletesResults.py read the podium table and visited each unique athlete profile URL to collect biographical fields (date of birth, height, weight, birth city, and affiliations) into the athletes table, and individual medal-winning results (year, sport, event, medal type, and NOC country code) into the results table. Country-level economic data were collected separately by countries.py, which scraped GDP from worldometers.info/gdp/gdp-by-country/ and population from worldometers.info/world-population/population-by-country/. Both pages were parsed using CSS class and table-element selectors robust to minor layout variation. GDP per capita was computed as GDP divided by population for each country. Full country names were mapped to three-letter Olympic NOC codes using a manually constructed dictionary derived from Statista's Winter Olympics medal-country reference; defunct states were removed, leaving 39 active countries. Edge cases such as "United Kingdom" versus "Great Britain" were resolved manually within the mapping. The resulting countries table was stored in the same database.

For analysis, the four tables were linked in analysis.py. The results table was joined to athletes on athlete_url (left join) and then to countries matching results.country to countries.country_code (left join), producing a single flat DataFrame of all individual medal records enriched with athlete biographical data and country economic indicators.

Politeness constraints were applied throughout. Requests to Olympedia.org were separated by a 15-second delay to exceed the site's 10-second crawl policy safely; requests to Worldometers.info were separated by a 20-second delay, exceeding the 0-second requirement of the site. All requests included a descriptive User-Agent header identifying the project and a contact email address.

B. *CLEANING DATA*

After merging, the height and weight columns were coerced to numeric types using pandas; values that could not be parsed were set to NaN and excluded from any calculation requiring those fields. The dob column was parsed as datetime with errors='coerce', so malformed or missing birth dates became NaT. Athlete age at competition was derived by subtracting the athlete's birth year from the Olympic year of the result record.

Several structural quality issues were identified and resolved during acquisition. Athletes who tied in a competition were displayed on Olympedia with placement labels such as "=1" or "=2" rather than a clear medal designation; this was resolved by detecting medal-coloured span elements (Gold, Silver, Bronze) directly in the HTML rather than relying on placement rank. Athletes who competed for different countries across Games would have produced incorrect joins if countries were stored on the athletes table; this was corrected by recording the NOC code on each individual result row in the results table, making it the sole join key to the countries table.

Relay and team event rows were excluded during scraping: the athletesResults.py parser checked whether the team-cell in each result row was non-empty and skipped those rows, retaining only individual-event medal records. Duplicate athlete entries arising from athletes appearing on multiple edition pages were removed from the podium table by deduplicating on athleteURL and retaining the last occurrence. For the body-profile analysis (Figure 3), sports with fewer than 15 medalists having complete height, weight, and age records were excluded to avoid unreliable means.

C. *ANALYSIS APPROACH*

Twenty-five visualizations were produced from the integrated dataset to address the three research sub-questions.

Figures addressing national wealth and medal success were spread across multiple chart types. Figure 1 juxtaposed GDP per capita and total medal count for the top eight medal-winning countries using a dual-axis grouped bar chart. Figure 5 extended this to the top 15 countries by stacking gold, silver, and bronze counts in a single horizontal bar chart, making medal composition visible alongside volume. Figure 6 plotted log-transformed national GDP against total medals as a bubble scatter chart (bubble size scaled by GDP per capita), with a linear trend line and Pearson r annotation. Figure 7 ranked the 12 most efficient nations by medals per million population as a horizontal bar chart, annotating each bar with its raw medal count. Figure 11 repeated the scatter approach using log-transformed population in place of GDP. Figure 12 presented a four-panel 2x2 dashboard comparing log GDP, GDP per capita, log population, and GDP per capita against medals per million population, each panel annotated with its correlation coefficient. Figures 13 and 14 quantified predictive power more formally: Figure 13 displayed R^2 values for four country-level factors across each eligible sport as a heatmap (sports with fewer than eight represented countries excluded), while Figure 14 rendered the same values as a grouped bar chart for direct cross-factor comparison within each sport. Figure 21 ranked the top 15 countries by gold medal share as a horizontal bar chart. Figure 23 plotted GDP per capita against medals per million people to surface countries that overperform relative to their wealth, with the five most efficient nations labelled. Figure 24 computed the per-sport difference between mean medalist height for the top eight countries and the global average, displayed as a diverging horizontal bar chart.

Figures addressing temporal trends used area and line-based encodings. Figure 2 used a stacked area chart to show how the share of medals among the top ten countries shifted across the five Games from 2010 to 2026, with Canada highlighted in a distinct colour. Figure 8 presented medal trajectories for the top eight countries as a 2x4 small-multiple grid, overlaying total medal count and gold medal count with filled areas for each. Figure 25 identified the 12 fastest-improving countries by computing the change in medal count between the first and last Games and displaying the result as a horizontal bar chart.

Figures addressing sport structure and medal distribution used bar and scatter encodings. Figure 4 plotted total medals awarded per sport as a horizontal bar chart with bars colour-coded by broad category (skiing/biathlon/Nordic, skating, and other). Figure 20 plotted total medals against the number of competing countries per sport as a bubble scatter chart, with bubble size proportional to the ratio of medals to countries. Figure 22 ranked sports by the share of medals captured by the top three countries, surfacing which disciplines are most dominated by a small number of nations.

Figures addressing physical traits of medalists used distributional and profile-based encodings. Figure 3 displayed mean height, weight, and age for each eligible sport side by side as a three-panel horizontal dot plot with connecting lines from a fixed baseline, restricted to sports with at least 15 medalists with complete measurements. Figures 9, 15, and 16 showed the full distribution of height, weight, and age respectively for each sport using overlaid box plots and strip plots ordered by median value, restricted to sports with at least 10 medalists with valid measurements. Figure 18 plotted year-on-year trends in mean height, weight, and age across three panels, drawing a faint line per sport and a bold highlighted series for the cross-sport average. Figure 19 decomposed the same trends into a sport-by-sport small-multiple grid with y-axes zoomed to actual variation. Figure 10 displayed each country's proportion of gold, silver, and bronze medals as a three-panel heatmap, restricted to countries with at least 10 total medals. Figure 17 ranked the 15 highest medal-producing cities by sport as a horizontal bar chart, with bars colour-coded by sport. All

figures were created with Matplotlib and Seaborn and saved as PNG files in the figures/ directory.

4. FINDINGS (TABLES, VISUALIZATION, KEY OBSERVATIONS) & DISCUSSIONS

A. *RELATIONSHIP BETWEEN NATIONAL FACTORS AND MEDAL EFFICIENCY*

While there is no clear or strong relationship observed between total GDP and total medal count (correlation score r=0.55) or GDP per capita and total medal count (r=0.25), GDP per capita is strongly related to medal efficiency, which indicates total medals per each million population, or so-called medals per capita. Their correlation score is r = 0.74. We also examined the relationship between country population and total medals, but their correlation is weak (r=0.34). In other words, a bigger population does not translate to a larger number of medals, and economic strength does not predict total medal count but plays a role in medal efficiency. The chart below shows all the above-mentioned relationships.

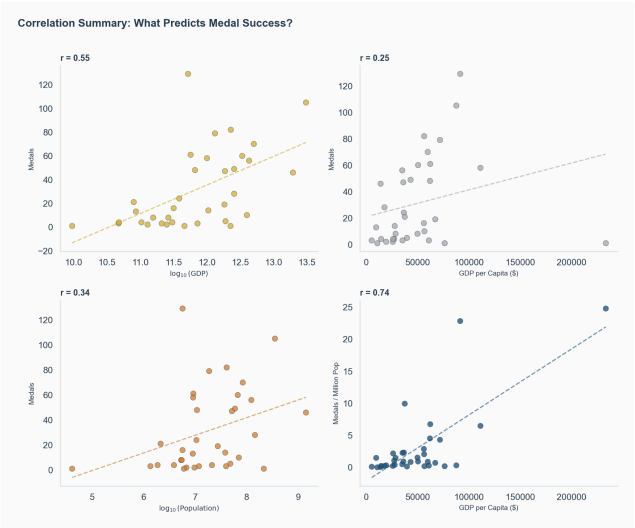

Figure 1: Correlation between medal success and national factors at Winter Olympics (2010-2026)

Small population countries such as Norway, Slovenia, Austria, and Switzerland demonstrate high medal efficiency relative to GDP per capita. Canada has a relatively high GDP per capita but lower medal efficiency compared to top-performing nations. There is room for improvement for Canada to improve on this metric.

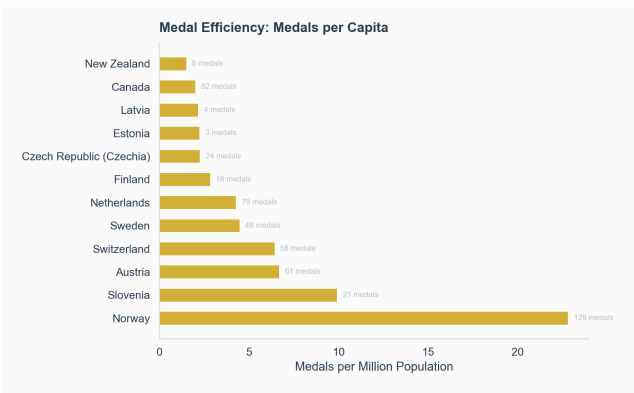

Figure 2: Media Efficiency by top-performing countries at the Winter Olympics (2010-2026)

B. *SPORT OPPORTUNITIES AND MEDAL DISTRIBUTION*

More than 80% of all Olympic medals are offered by only two sports. Skiing disciplines account for the largest share of medal opportunities (56%), followed by skating (26%).

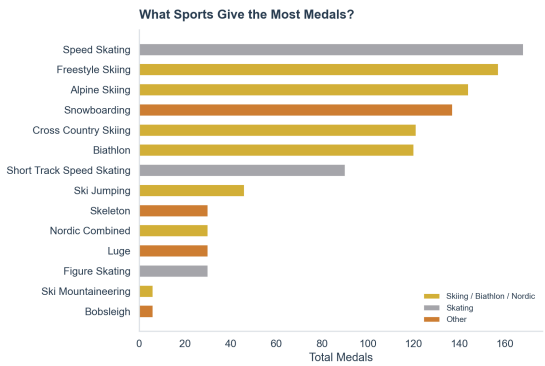

Figure 3: Disciplines that offer the most medals at Winter Olympics (2010-2026)

From our data queries, though Canada's medal performance is concentrated in long track speed skating, short track speed skating, freestyle skiing, figure skating, and

snowboarding, they are not leading in these sports. Also, Canada has demonstrated broad participation. They attended 14 disciplines in the Winter Olympics 2026. In contrast, Norway participated in fewer disciplines (10) yet achieved higher total medal counts. This performance is largely associated with their leading sports in cross-country skiing, biathlon, Nordic combined, and ski jumping, in addition to competitive performance in speed skating.

The following chart shows the top high-medal sports and the competitiveness (number of competing countries). The main medal-producing sports of Canada are also among the most competitive ones (Freestyle skiing and snowboarding). Norway's medal counts come from the mid-range level of competitive sports such as cross-country skiing and Biathlon.

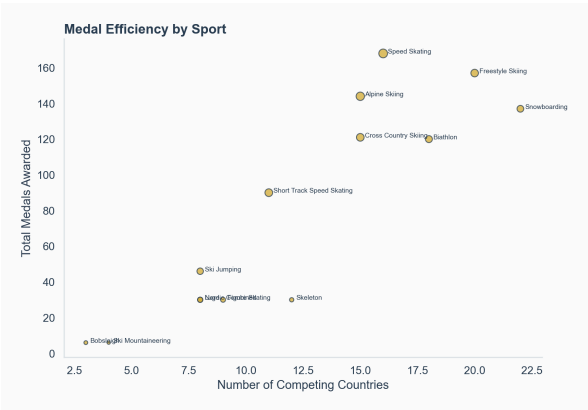

Figure 4: Media Success and Competing Country Count at Winter Olympics (2010-2026)

An implication for Canada is that, instead of broad participation, more focus on sports with both high medal availability and moderate competition intensity, such as short track speed skating and long track speed skating, may offer a strategic advantage.

C. *MEDALIST CHARACTERISTICS (AGE, WEIGHT, HEIGHT)*

Medalist profiles vary significantly across sports, indicating that there is no single set of physical characteristics associated with success in the Winter Olympics. Weight and age appear to be more distinguishing factors than height. Lighter athletes tend to dominate sports such as ski jumping and figure skating, while heavier athletes overrepresent in sliding sports such as luge and skeleton. Athletes with moderate body builds perform well in skiing, snowboarding, and skating disciplines.

It is worth observing that athletes in their early 30s remain competitive in certain sports. In other words, peak performance age depends on the disciplines.

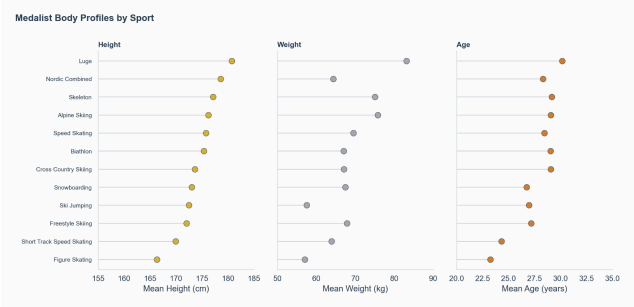

Figure 5: Medalist Characteristics (Height, Weight, and Age) by Sport at Winter Olympics (2010-2026)

The findings suggest that reaching the podium is sport-specific and indicate that the universal profile of medalists across disciplines may not exist. Better alignments between athletic profiles and the sports where their physical characteristics give them natural advantages to medal success can be an opportunity for Canada. Doing so does not suggest forcing athletes or potential people into certain sports but presents them with opportunities in which they are able to utilize their strength. The age characteristic also implies training development timeframe is sport-specific to avoid prematurely dropping athletes that may have peak performance later in their careers.

5. LIMITATIONS AND FUTURE WORK

A. *LIMITATIONS*

Data quality and data acquisition
Though the dataset has considerable records (1150 rows), the data for this project is entirely acquired through web scraping. It results in missing or inconsistent values for some variables. Also, some records were skipped due to page unavailability or scraping errors. That affects the data robustness and reliability.

Lack of explanatory variables
In our initial plan, two of the potential predictors were funding allocation for athletes and athletic years of experience. While in theory, the latter is feasible to collect through web scraping. In practice, it is not due to time constraints. Regarding the variable of funding allocation, it is challenging to obtain due to different country funding systems for their national teams and the complexity of commercial sponsorship. We used the variable of GDP or GDP per capita instead, which is less direct to explain the study subject.

This study also lacks other potential factors, such as coaching investment, training infrastructure investment, or qualitative factors, such as national policy towards talent development or sport participation for citizens, among the top-performing countries. They are important to bringing a more holistic view than just examining quantitative data.

B. *FUTURE WORK*

Data quality and data acquisition
Future work should incorporate more robust data acquisition strategies to handle errors and acquire more structured data using other techniques to increase data robustness and data quality.

Mixed-method approach
Combining quantitative and qualitative methods would enable comprehensive insights to predict medal-winning opportunities at the Winter Olympics.

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